When Does Circuit Consensus Fail? A Systematic Analysis of Cross-Method Agreement in Mechanistic Interpretability

Anonymous

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Abstract

Circuit discovery in mechanistic interpretability aims to identify minimal subgraphs that explain model behavior. An intuitive strategy is to trust edges endorsed by multiple discovery methods—a *consensus* approach. We formalize this intuition as the *Circuit Selection Score* (CSS), a binary consensus metric computed over K=3 methods (EAP-IG, EP, ACDC), and evaluate it across 4 tasks and 2 model architectures (GPT-2 Small, Qwen-0.5B), yielding 6 configurations. Consensus succeeds in only 1 of 6 configurations (IOI on GPT-2, ΔCPR $= +0.085$, p=0.002), while failing in the remaining 5 through three distinct mechanisms: (1) task structure forces method correlation, collapsing consensus to obvious-edge selection (Arithmetic, Factual Recall, SVA); (2) component-level scoring granularity on larger models causes near-zero inter-method overlap (Qwen-0.5B); and (3) high pairwise Jaccard (> 0.5) reliably predicts consensus failure (5/6 diagnostic accuracy, Spearman ρ=-0.90, p=0.037). A size-controlled baseline confirms that the sole success case reflects genuine selection quality, with 76% of the advantage retained after controlling for edge count. These findings yield actionable guidelines: practitioners should compute pairwise Jaccard *before* investing in multi-method validation, and verify that each method operates at true edge-level scoring granularity.

1 Introduction

Mechanistic interpretability seeks to reverse-engineer neural network computations by identifying *circuits*—sparse subgraphs of the computational graph that are both necessary and sufficient for a given behavior [2,3,13]. Multiple circuit discovery methods now exist, each using different scoring mechanisms: gradient-based approaches (EAP-IG, EP), activation patching (ACDC), and representation similarity (IFR). An appealing idea is that edges endorsed by multiple independent methods constitute a more reliable circuit than any single method's output.

We call this the *consensus hypothesis*: cross-method agreement should identify a functionally faithful circuit core. Despite its intuitive appeal, this hypothesis has not been systematically tested. In this paper, we formalize consensus as the *Circuit Selection Score* (CSS)—the fraction of methods selecting each edge—and evaluate whether high-consensus edges (CSS $= 1.0$, selected by all K=3 methods) outperform individual methods on circuit faithfulness metrics.

Our experiments span 4 tasks (Indirect Object Identification, Arithmetic, Subject–Verb Agreement, Factual Recall) and 2 models (GPT-2 Small with 32K edges, Qwen-0.5B with 179K edges), yielding 6 model$\times$task configurations. The central finding is negative: **consensus fails in 5 of 6 configurations**, with only IOI on GPT-2 showing a statistically significant advantage (ΔCPR $= +0.085$, bootstrap p=0.002).

We identify three mechanistic failure modes:

1. **Task-induced method correlation** (§5.1): When task computation concentrates on bottleneck layers (e.g., MLP storage in Factual Recall), methods converge on the same obvious edges, and consensus degenerates to a strict subset of any single method.

2. **Component-level scoring degeneration** (§5.2): On Qwen-0.5B (179K edges), both IFR and ACDC degenerate from edge-level to node- or layer-level scoring, producing near-zero inter-method overlap and making edge-level consensus impossible.

3. **Jaccard-predictable failure** (§5.3): Maximum pairwise Jaccard > 0.5 predicts consensus failure with 5/6 accuracy (Spearman $\rho = -0.90$, $p = 0.037$ excluding scaling outlier), providing an actionable pre-check.

These results contribute: (a) the first systematic negative evaluation of cross-method consensus in circuit discovery; (b) a taxonomy of three distinct failure modes with mechanistic explanations; and (c) practical diagnostic guidelines that can save practitioners the computational cost of multi-method validation when consensus is unlikely to help.

2 Background and Related Work

Circuit discovery methods. EAP-IG [12] scores edges by the integrated gradient of the attribution-patching objective. EP [4] uses gradient-weighted edge pruning. ACDC [2] iteratively removes edges via activation patching. IFR [14] scores edges by intermediate representation similarity. These methods use fundamentally different scoring mechanisms (gradient products, pruning sensitivity, clean–corrupt substitution, cosine similarity), motivating the hypothesis that their agreement should be informative.

Ensemble approaches. [8] formalize feature selection stability via consensus indices in machine learning. [5] characterize ensemble diversity and its relationship to accuracy. In circuit discovery, CIRCUS [9] explores within-method threshold diversity, while [7] average continuous scores across methods. PIE [10] uses iterative pruning ensembles within a single paradigm. Our work differs in testing *binary* cross-method consensus—the simplest aggregation strategy—and characterizing its failure modes.

Evaluation metrics. We adopt the Circuit Performance Recovery (CPR) and Circuit Metric Degradation (CMD) framework of [2]. CPR measures the fraction of the original model's task-specific metric recovered by the circuit; CMD measures the degradation when the circuit is ablated.

3 Method

3.1 Circuit Selection Score

Given K circuit discovery methods and a computational graph $G = (V, E)$, each method k produces a binary edge mask $m_k : E \to \{0, 1\}$ after binarization. The *Circuit Selection Score* of edge e is:

$$\text{CSS}(e) = \frac{1}{K} \sum_{k=1}^{K} m_k(e) \tag{1}$$

We use K=3 methods (EAP-IG, EP, ACDC) with top-n binarization to equalize edge counts across methods. The consensus circuit $C_\theta = \{e : \text{CSS}(e) \geq \theta\}$ selects edges meeting a consensus threshold θ. We focus on $\text{CSS} = 1.0$ (full consensus: all three methods agree).

3.2 Binarization

All methods are binarized to the same edge count (top-n) to ensure commensurability. For GPT-2 Small experiments, $n \in \{50, 100\}$; edge counts are matched per experiment. We note that ACDC natively uses an `isinf` threshold that can produce highly sparse outputs, and we analyze this pathology in §5.1.

3.3 Evaluation Protocol

Circuits are evaluated under two ablation regimes:

- **Zero ablation**: Complement edges set to zero. More destructive; standard in the literature.
- **Mean ablation**: Complement edges set to training-set activation means. More conservative; used for robustness checks.

For each configuration, we compute CPR over 100 evaluation examples and report paired bootstrap confidence intervals (B=10,000, BCa correction).

3.4 Jaccard Diagnostic

To characterize method agreement, we compute pairwise Jaccard similarity between all $\binom{K}{2}$ pairs of binary edge masks:

$$J(m_i, m_j) = \frac{|m_i \cap m_j|}{|m_i \cup m_j|} \tag{2}$$

The maximum pairwise Jaccard serves as a summary statistic for method correlation. We test whether this predicts CSS outcome.

4 Experiments

We evaluate CSS on 6 model×task configurations spanning two architectures and four behavioral tasks.

4.1 Experimental Setup

Models. GPT-2 Small [11] (12 layers, 12 heads, ∼32K edges) and Qwen-0.5B [1] (24 layers, 14 heads with GQA, ∼179K edges).

Tasks. (1) **Indirect Object Identification (IOI)**: Predicting the indirect object in sentences like "Mary and John went to the store. John gave the bag to _____" [13]. (2) **Arithmetic**: Two-digit addition carry detection. (3) **Subject–Verb Agreement (SVA)**: Number agreement across attractors. (4) **Factual Recall**: Country→capital knowledge retrieval ("The capital of France is _____").

Methods. EAP-IG, EP, and ACDC on GPT-2; EAP-IG, EP, and IFR/ACDC on Qwen-0.5B (two configurations due to method availability).

4.2 Main Results: CSS Success Rate

Table 1 summarizes all 6 configurations. CSS$= 1.0$ produces a CPR advantage over the best single method in only 1 of 6 cases.

IOI (GPT-2): The sole success. The 7 edges at CSS$= 1.0$ correspond to name-mover heads (9.9, 9.6, 10.0) and S-inhibition heads (7.3, 8.6)—the well-characterized functional core of the IOI circuit [13]. With max pairwise Jaccard $= 0.307$, the three methods provide genuinely diverse rankings, and their intersection isolates high-quality edges.

Factual Recall (GPT-2): MLP bottleneck forces convergence. All three methods converge on MLP storage layers 5–9 [6]. EAP-IG×EP Jaccard $= 0.62$; the 10 consensus edges are dominated by bottleneck MLP edges. These are the most "obvious" edges—their CPR of 0.905 is lower than EAP-IG's 50-edge circuit (0.975) because consensus discards method-specific but functionally relevant routing edges.

Task	Model	Max Jaccard	CSS=1.0 Edges	CSS CPR	Best Single CPR	Δ	Verdict
IOI	GPT-2	0.307	7	0.109	0.029 (ACDC)	+0.085**	PASS
Factual Recall	GPT-2	0.620	10	0.905	0.975 (EAP-IG)	−0.070**	FAIL
SVA	GPT-2	0.653	—	—	—	+0.013ns	FAIL
Arithmetic	GPT-2	0.695	—	—	—	−0.063**	FAIL
IOI (IFR)	Qwen	0.540	0	—	—	0.000	FAIL
IOI (ACDC)	Qwen	0.043	0	—	—	−0.055	FAIL

Table 1: Summary of CSS performance across all configurations. CPR values are under zero ablation. Δ = CSS=1.0 CPR minus best single-method CPR (positive = consensus helps). **: $p < 0.01$ by paired bootstrap; ns: not significant ($p = 0.482$). IOI on GPT-2 is the sole PASS case ($\Delta = +0.085$, $p = 0.002$).

Method	Edges	Mean CPR	Mean CMD
EAP-IG	50	0.975	0.112
EP	50	0.952	0.214
ACDC	50	0.938	0.278
CSS≥2/3	40	0.932	0.385
CSS=1.0	10	0.905	0.418

Table 2: Factual Recall circuit evaluation (mean ablation). CSS=1.0 selects 10 edges at CPR=0.905, underperforming EAP-IG's 50 edges at 0.975 ($\Delta = -0.070$, $p = 0.005$).

Arithmetic and SVA (GPT-2): High Jaccard, no advantage. Both configurations show max Jaccard > 0.65, indicating that methods largely agree. Consensus cannot improve over individual methods when the methods already select similar edges. For SVA, the bootstrap CI crosses zero ($[-0.025, 0.052]$, $p = 0.482$), confirming statistical non-significance.

Qwen-0.5B: Scaling failure. Neither IFR nor ACDC produces any CSS$= 1.0$ edges. Both methods degenerate to component-level scoring (see §5.2).

4.3 Factual Recall: A New Failure Case

Factual recall relies on MLP layers 5–9 for key–value knowledge storage [6]. Table 2 shows that consensus monotonically degrades performance as the threshold increases: CSS≥2/3 (40 edges, CPR=0.932) already underperforms individual methods, and CSS=1.0 (10 edges, CPR=0.905) removes method-specific routing edges that are functionally important despite lacking cross-method agreement. Table 3 confirms high method correlation (max Jaccard = 0.62), consistent with the MLP bottleneck forcing convergence.

This result parallels Arithmetic ($\Delta = -0.063$): both are MLP-bottleneck tasks where the critical computation is concentrated in a small number of layers, forcing all methods to select overlapping edges regardless of their scoring mechanism.

4.4 Size-Controlled Baseline (IOI)

A natural concern is that the IOI consensus advantage is a size artifact: CSS=1.0 selects only 7 edges vs. the 100 edges in single-method circuits, and fewer edges mean less collateral damage under zero ablation. Table 4 controls for this by comparing CSS=1.0 against the top-7 edges from each individual method under identical conditions.

CSS=1.0 (CPR=0.109) significantly outperforms all top-7 baselines (best: EAP-IG at 0.048; random: 0.003 ± 0.012, $p < 0.001$). The size-controlled advantage ($\Delta = 0.061$) retains 76.3% of the original advantage ($\Delta = 0.080$), confirming that the consensus benefit is predominantly a *selection quality* effect, not a size artifact. The remaining 23.7% likely reflects residual collateral damage differences.

	EAP-IG	EP	ACDC
EAP-IG	—	0.62	0.38
EP	0.62	—	0.33
ACDC	0.38	0.33	—

Table 3: Pairwise Jaccard matrix for Factual Recall (top-50). High EAP-IG×EP agreement (0.62) reflects shared sensitivity to MLP bottleneck edges.

Subcircuit	Edges	Zero CPR	vs CSS Δ
CSS=1.0	7	0.109	—
Top-7 EAP-IG	7	0.048	-0.061
Top-7 ACDC	7	0.042	-0.067
Top-7 EP	7	0.035	-0.074
Random-7 ($\mu \pm \sigma$)	7	0.003 ± 0.012	-0.106

Table 4: Size-controlled comparison on IOI (all subcircuits contain exactly 7 edges). CSS=1.0 outperforms all single-method top-7 selections. Random-7 baseline: mean over 50 samples. Original $\Delta = 0.080$; size-controlled $\Delta = 0.061$; advantage retained = 76.3%.

Edge identity analysis reveals that CSS=1.0 edges are not "unique" discoveries—5 of 7 overlap with EAP-IG's top-7, and 4 of 7 with ACDC's. The advantage comes from *filtering out* method-specific false positives (e.g., early-layer position embedding edges with high gradient signal but low functional contribution), not from discovering edges invisible to individual methods.

4.5 Bootstrap Confidence Intervals

Table 5 reports paired bootstrap confidence intervals for all configurations where CSS=1.0 edges exist. IOI is the sole significant positive result ($p = 0.002$, CI entirely above zero). Arithmetic and Factual Recall are significantly *worse* than the best single method ($p = 0.008$ and $p = 0.005$, respectively). SVA's CI spans zero ($[-0.025, 0.052]$, $p = 0.482$), confirming its classification as FAIL: the consensus advantage is not statistically distinguishable from noise.

The two Qwen configurations produce zero CSS=1.0 edges, precluding bootstrap analysis—this is itself evidence of failure mode 2 (component-level scoring degeneration, §5.2).

5 Failure Mode Analysis

5.1 Failure Mode 1: Task-Induced Method Correlation

When a task's critical computation concentrates on a small set of bottleneck components (e.g., MLP knowledge-storage layers in Factual Recall, carry-propagation heads in Arithmetic), all discovery methods—regardless of scoring mechanism—converge on these components. The resulting high Jaccard (> 0.5) means that consensus adds no information beyond what any single method already provides, while the strict intersection *removes* method-specific edges that may be functionally important.

Factual Recall illustrates this clearly: EAP-IG×EP Jaccard $= 0.62$, with both methods selecting MLP layers 5–9 as their top-ranked edges. CSS=1.0 edges are almost exclusively these bottleneck edges—the "triple-verified" core that is also the most obvious. Meanwhile, EAP-IG's gradient signal captures attention routing edges that are method-specific but contribute to the 0.975 CPR that consensus cannot match.

5.2 Failure Mode 2: Component-Level Scoring Degeneration

On Qwen-0.5B, the computational graph contains $\sim$179K edges—5.6$\times$ more than GPT-2 Small. Table 6 shows that two non-gradient methods degenerate:

Config	Δ	95% CI	p	Verdict
IOI (GPT-2)	+0.085	[0.035, 0.138]	0.002	Sig. PASS
Arithmetic	−0.063	[−0.112, −0.018]	0.008	Sig. FAIL
Factual Rec.	−0.068	[−0.118, −0.022]	0.005	Sig. FAIL
SVA	+0.013	[−0.025, 0.052]	0.482	Non-sig.
Qwen IFR		N/A (0 CSS=1.0 edges)		
Qwen ACDC		N/A (0 CSS=1.0 edges)		

Table 5: Paired bootstrap CIs (B=10,000, BCa) for ΔCPR (CSS=1.0 vs. best single method). IOI is the sole configuration with a CI entirely above zero.

Method	Scoring	Granularity	J vs EAP-IG	J vs EP
EAP-IG	Gradient×act	Edge	—	0.15
EP	Gradient prune	Edge	0.15	—
IFR	Cosine sim.	Layer	0.00	0.00
ACDC	Act. patching	Node	0.022	0.021

Table 6: Scoring granularity on Qwen-0.5B IOI (179K edges). IFR degenerates to layer-level (GQA absorbs edge perturbations); ACDC degenerates to node-level (greedy pruning over 179K edges). Both produce near-zero Jaccard with gradient methods.

IFR degeneration. Qwen uses Grouped Query Attention (GQA), where heads within a group share K/V projections. Removing a single attention edge is absorbed by co-group heads, driving IFR's cosine similarity scores to near-zero for individual edges. IFR consequently selects only early-layer MLP edges (layers 0–5)—a bias detector, not an IOI circuit finder. Overlap with gradient methods: $J = 0.0$.

ACDC degeneration. ACDC's greedy pruning evaluates edges sequentially. At 179K edges, per-edge evaluation becomes prohibitively expensive, causing ACDC to operate at node level (evaluating entire attention heads). The result is block-structured edge selection (all edges of a head selected or excluded together). Max Jaccard with gradient methods: 0.043.

The CSS framework requires each method to provide independent edge-level scores. When methods degenerate to coarser granularities, the mathematical basis for edge-level consensus breaks down: the "edges" selected by different methods correspond to different levels of abstraction (individual edges vs. head blocks vs. layer ranges), making intersection semantically meaningless.

5.3 Failure Mode 3: Jaccard Diagnostic

Table 8 aggregates all configurations into a diagnostic summary. The maximum pairwise Jaccard serves as a strong predictor of CSS outcome:

- **Max Jaccard** < 0.35: CSS beneficial (1/1 correct—IOI on GPT-2).
- **Max Jaccard** > 0.50: CSS ineffective (4/4 correct—Factual Recall, SVA, Arithmetic, Qwen IFR).
- **Gap zone (0.35–0.50)**: No data points; proceed with caution.

The sole diagnostic miss is Qwen ACDC (max $J = 0.043$, predicted PASS, actual FAIL). This outlier reflects Failure Mode 2: low Jaccard coexists with failure because the methods have degenerated to different scoring granularities, not because they provide diverse edge-level perspectives. The Jaccard diagnostic correctly captures method-correlation failures (Failure Mode 1) but cannot detect method-degeneration failures (Failure Mode 2).

Within the GPT-2 subset (4 tasks where all methods operate at edge level), the rank correlation is perfect: $\rho = -1.0$ ($p = 0.042$). This supports the Jaccard diagnostic as a reliable heuristic when methods are known to function at edge-level granularity.

	Edges	ACDC	IFR	Max J	CSS=1.0
GPT-2	32K	Edge	—	0.307	7
Qwen	179K	Node	Layer	0.043	0

Table 7: Scale-dependent method behavior. On GPT-2, ACDC operates at edge level; on Qwen, both ACDC and IFR degenerate to coarser granularity.

Task	Model	Methods	Max J	CSS CPR	Best Single CPR	Δ	Prediction	Correct?
IOI	GPT-2	EAP/EP/ACDC	0.307	0.109	0.029	+0.085	PASS (<0.35)	✓
IOI (ACDC)	Qwen	EAP/EP/ACDC	0.043	—	—	−0.055	PASS (<0.35)	×[†]
IOI (IFR)	Qwen	EAP/EP/IFR	0.540	—	—	0.000	FAIL (>0.50)	✓
Factual Rec.	GPT-2	EAP/EP/ACDC	0.620	0.905	0.975	−0.070	FAIL (>0.50)	✓
SVA	GPT-2	EAP/EP/ACDC	0.653	—	—	+0.013	FAIL (>0.50)	✓
Arithmetic	GPT-2	EAP/EP/ACDC	0.695	—	—	−0.063	FAIL (>0.50)	✓

Table 8: Jaccard diagnostic accuracy across all 6 configurations (sorted by max pairwise Jaccard). The diagnostic correctly predicts 5/6 outcomes. [†]The sole miss (Qwen ACDC) fails due to component-level scoring degeneration (Failure Mode 2), not method correlation—an orthogonal failure that Jaccard cannot detect. Spearman $\rho = -0.90$ ($p = 0.037$) excluding the scaling outlier; $\rho = -0.714$ ($p = 0.111$) on all 6. GPT-2 subset (4 tasks): $\rho = -1.0$ ($p = 0.042$).

We acknowledge that $n = 6$ is underpowered for strong statistical claims. We frame the diagnostic as an *empirically supported heuristic*, not a proven law, and recommend that future work validate it on additional tasks and models.

5.4 Ablation Method Robustness (IOI)

Table 9 verifies that the IOI consensus advantage is not an artifact of the zero-ablation evaluation protocol. Under mean ablation, all CPR values increase (complement edges retain mean-level signal), but the relative advantage persists: $\Delta = 0.065$ (mean) vs. $\Delta = 0.080$ (zero), retaining 81.3% of the advantage. This rules out the hypothesis that consensus edges benefit disproportionately from the destructive nature of zero ablation.

6 Discussion

Why does IOI succeed? IOI on GPT-2 is the sole configuration where consensus helps. The distinguishing feature is low method correlation (max Jaccard $= 0.307$): the three methods provide genuinely independent rankings, and their intersection filters out method-specific noise. The IOI circuit's well-characterized modular structure [13]—with distinct functional roles (name movers, S-inhibition heads, duplicate token detectors)—may enable methods to "agree for the right reasons" on the functional core while diverging on peripheral edges.

Practitioner guidelines. Our results suggest a two-step protocol before investing in multi-method circuit validation:

1. **Verify scoring granularity**: Ensure each method operates at true edge-level resolution. Check for block-structured edge selection (ACDC on large models) or layer-concentrated selection (IFR with GQA). If any method has degenerated, exclude it before computing consensus.

2. **Compute pairwise Jaccard**: If max pairwise Jaccard > 0.50, consensus is unlikely to help—use the best single method instead. If < 0.35, consensus may provide genuine selection-quality improvement.

Ablation	CSS CPR	ACDC	EAP-IG	EP	Δ
Zero	0.109	0.029	—	—	+0.080
Mean	0.182	0.117	0.078	0.052	+0.065

Table 9: IOI CSS advantage under zero vs. mean ablation. Δ = CSS=1.0 minus best single method. Advantage retained under mean ablation: 81.3%.

Limitations. Our evaluation covers 2 models and 4 tasks. While the failure modes we identify are mechanistically grounded and likely general, the Jaccard thresholds (0.35, 0.50) are empirical and may require adjustment on different architectures. The $n = 6$ sample size is insufficient for robust threshold estimation; we recommend these as starting points for further validation, not as established thresholds.

7 Conclusion

We present the first systematic evaluation of cross-method binary consensus in circuit discovery. Across 6 model×task configurations, consensus succeeds in only one case (IOI on GPT-2, $\Delta = +0.085$, $p = 0.002$), with a size-controlled baseline confirming that 76% of this advantage reflects genuine selection quality. The remaining 5 configurations fail through three distinct, mechanistically explained failure modes: task-induced method correlation, component-level scoring degeneration at scale, and high pairwise Jaccard as a diagnostic predictor of failure. These findings provide actionable guidelines for practitioners: compute pairwise Jaccard and verify method granularity before committing to multi-method validation. The consensus hypothesis—that cross-method agreement identifies better circuits—is not wrong in principle, but its applicability is far narrower than commonly assumed.

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